

Experiment 9

Student Name: Prateek Verma
Branch: MCA (CC & DEVOPS)
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Title: Triggers in SQL.

1. Aim:

- a. To implement database triggers in PostgreSQL to automatically calculate values and enforce constraints during data insertion operations.

2. Software Requirements:

- a. PostgreSQL (Database Server)
- b. pgAdmin
- c. OS: Windows

3. Objective:

- a. Trigger Implementation: To understand how to create and use triggers in PostgreSQL.
- b. Automation: To automate calculation of total payable amount using triggers.
- c. Data Integrity: To enforce constraints using trigger conditions.
- d. Real-time Processing: To execute logic automatically before inserting data.

4. Procedure of the Practical:

- a. Start the system.
- b. Open pgAdmin.
- c. Create and select the database in which you want to perform the experiment.
- d. Establish connection to the database using Alt+Shift+Q.
- e. Run the following queries given in Experiment Steps (5).

5. Experiment Steps:

a. Creating trigger function

```
CREATE OR REPLACE FUNCTION CACULATE_PAYABLE_AMOUNT() RETURNS TRIGGER
AS
$$
BEGIN
    NEW.total_payable_amount:=NEW.perhour_salary*New.working_hours;

    IF NEW.total_payable_amount>25000 THEN
        RAISE EXCEPTION 'INVALID ENTRY BEACUSE PAYABLE AMOUNT CAN NOT BE
GREATER THAN 25000';
    END IF;

    RETURN NEW;
END;
$$ LANGUAGE PLPGSQL;
```

b. Creating trigger

```
CREATE OR REPLACE TRIGGER AUTOMATED_PAYABLE_AMOUNT_CALCULATION
BEFORE INSERT
ON employee
FOR EACH ROW
EXECUTE FUNCTION CACULATE_PAYABLE_AMOUNT()
```

c. Inserting valid data

```
INSERT INTO EMPLOYEE(EMP_ID, EMP_NAME,working_hours,perhour_salary)
VALUES (1, 'AKASH',10,1000)
```

	emp_id [PK] integer	emp_name character varying (50)	working_hours integer	perhour_salary numeric	total_payable_amount numeric
1	1	AKASH	10	1000	10000

d. Inserting invalid data

```
insert into employee(emp_id, emp_name, working_hours, perhour_salary)
values (2,'Ankush',8,100000)
```

Data Output	Messages	Notifications
ERROR: INVALID ENTRY BEACUSE PAYABLE AMOUNT CAN NOT BE GREATER THAN 25000 CONTEXT: PL/pgSQL function caculate_payable_amount() line 6 at RAISE SQL state: P0001		

6. Input/Output Details:

- a. The input for the experiment is the SQL queries mentioned in Experiment Steps (5).
- b. The output for each query is attached after the query itself.

7. Learning Outcome

- a. Create and use triggers in PostgreSQL.
- b. Use triggers for automation in SQL.
- c. Enforce constraints using trigger conditions.
- d. Real-time execution of database logic.